

This appendix gives a summary of basic definitions and results in matrix algebra which are used in this book. The presentation is restricted to important definitions and theorems, without examples or proofs. There are many books that have an appendix on matrix algebra; see, for instance, [Mardia, Kent, and Bibby \(1999\)](#) or [Rao, Toutenburg, Shalabh, and Heumann \(2008\)](#). More detailed expositions, including proofs, are in [Graybill \(1961\)](#), [Magnus and Neudecker \(2002\)](#), [Schott \(2005\)](#), and [Searle \(2006\)](#).

## A.1 Definition and Elementary Matrix Operations

### Definition A.1 Matrix

A real matrix of order  $n \times p$  (short:  $n \times p$ -matrix) is a rectangular array

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ a_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{np} \end{pmatrix}$$

of  $n \cdot p$  real numbers  $a_{ij}$ , arranged in  $n$  rows and  $p$  columns. We often write  $A = (a_{ij})$ ,  $i = 1, \dots, n$ ,  $j = 1, \dots, p$ .

### Definition A.2 The Transpose of a Matrix

The transpose  $A'$  of a matrix  $A$  is formed by interchanging rows and columns:

$$A' = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ a_{1p} & a_{2p} & \cdots & a_{np} \end{pmatrix}.$$

By definition, we have  $(A')' = A$ .

### Definition A.3 Column Vector, Row Vector

A matrix with column order  $p = 1$  is called a column vector. Thus,

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

is a column vector with  $n$  components  $a_1, a_2, \dots, a_n$ .

Row vectors are written as column vectors transposed, i.e.,

$$\mathbf{a}' = (a_1, a_2, \dots, a_n).$$

Thus, an  $n \times p$ -matrix has  $n$  row vectors and  $p$  column vectors.

### Definition A.4 Square Matrix

A matrix  $A$  is a square matrix if it is of order  $n \times n$ . Its diagonal, with elements  $a_{11}, \dots, a_{nn}$ , is sometimes called main diagonal.

### Definition A.5 Diagonal Matrix

A square matrix  $D$  is a diagonal matrix if the elements below and above the main diagonal are zero. Thus,

$$D = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & \cdots & d_n \end{pmatrix}$$

or  $D = \text{diag}(d_1, \dots, d_n)$  in short notation.

### Definition A.6 Identity Matrix

The  $n \times n$ -diagonal matrix

$$\mathbf{I}_n = \text{diag}(1, \dots, 1) = \begin{pmatrix} 1 & 0 & \dots & 0 \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ 0 & \dots & \dots & 1 \end{pmatrix}$$

is called the identity matrix of order  $n \times n$ . In situations where no confusion can arise, we also drop the index  $n$  and simply write  $\mathbf{I}$ .

### Definition A.7 Symmetric Matrix

A square matrix  $\mathbf{A}$  is called symmetric if  $\mathbf{A} = \mathbf{A}'$ .

### Definition A.8 Sums and Scalar Multiplication of Matrices

The sum  $\mathbf{A} + \mathbf{B}$  of two  $n \times p$ -matrices  $\mathbf{A} = (a_{ij})$  and  $\mathbf{B} = (b_{ij})$  is defined as

$$\mathbf{A} + \mathbf{B} = (a_{ij} + b_{ij}).$$

The multiplication of  $\mathbf{A}$  with a scalar  $\lambda \in \mathbb{R}$  is defined as

$$\lambda \mathbf{A} = (\lambda a_{ij}).$$

### Theorem A.1 Rules for Sums and Scalar Multiplication

Let  $\mathbf{A}, \mathbf{B}, \mathbf{C}$  be  $n \times p$ -matrices, and let  $r, k \in \mathbb{R}$  be scalars. We then have the following rules:

1.  $\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$
2.  $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
3.  $(k + r)\mathbf{A} = k\mathbf{A} + r\mathbf{A}$  and  $k(\mathbf{A} + \mathbf{B}) = k\mathbf{A} + k\mathbf{B}$
4.  $(kr)\mathbf{A} = k(r\mathbf{A})$
5.  $(k\mathbf{A})' = k\mathbf{A}'$
6.  $(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'$

### Definition A.9 Multiplication of Matrices

The product of the  $n \times p$ -matrix  $\mathbf{A} = (a_{ij})$  with the  $p \times m$ -matrix  $\mathbf{B} = (b_{ij})$  is the  $n \times m$ -matrix

$$\mathbf{AB} = \mathbf{C} = (c_{ik}) \quad \text{where} \quad c_{ik} = \sum_{j=1}^p a_{ij}b_{jk}.$$

Thus,

$$\mathbf{A} \cdot \mathbf{B} = \begin{pmatrix} \sum_{j=1}^p a_{1j} b_{j1} & \cdots & \sum_{j=1}^p a_{1j} b_{jm} \\ \vdots & \ddots & \vdots \\ \sum_{j=1}^p a_{nj} b_{j1} & \cdots & \sum_{j=1}^p a_{nj} b_{jm} \end{pmatrix}.$$

Note that two matrices  $\mathbf{A}$  and  $\mathbf{B}$  can only be multiplied if the number of columns of  $\mathbf{A}$  equals the number of rows of  $\mathbf{B}$ . In general, multiplication of two matrices is not commutative, i.e., in general  $\mathbf{B} \cdot \mathbf{A} \neq \mathbf{A} \cdot \mathbf{B}$ .

### Theorem A.2 Representing Sums as Products of Vectors

Let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  be column vectors and let  $\mathbf{1}$  be the  $n \times 1$ -(column) vector  $(1, \dots, 1)'$ . It follows:

1.  $\sum_{i=1}^n x_i = \mathbf{1}'\mathbf{x} = \mathbf{x}'\mathbf{1}$
2. Scalar product of  $\mathbf{x}$  and  $\mathbf{y}$ :  $\sum_{i=1}^n x_i y_i = \mathbf{x}'\mathbf{y} = \mathbf{y}'\mathbf{x}$
3. Squared Euclidean length of  $\mathbf{x}$ :  $\sum_{i=1}^n x_i^2 = \mathbf{x}'\mathbf{x}$

### Theorem A.3 Rules for Multiplication of Matrices

Let  $\mathbf{A}$ ,  $\mathbf{B}$ , and  $\mathbf{C}$  be matrices of appropriate dimensions. We then have the following rules:

1.  $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{A}\mathbf{B} + \mathbf{A}\mathbf{C}$
2.  $(\mathbf{A}\mathbf{B})\mathbf{C} = \mathbf{A}(\mathbf{B}\mathbf{C})$
3.  $(\mathbf{A}\mathbf{B})' = \mathbf{B}'\mathbf{A}'$
4.  $\mathbf{A}\mathbf{I} = \mathbf{A}$  and  $\mathbf{I}\mathbf{A} = \mathbf{A}$

### Definition A.10 Kronecker Product

The Kronecker product  $\mathbf{A} \otimes \mathbf{B}$  of the  $n \times p$ -matrix  $\mathbf{A}$  and the  $r \times q$ -matrix  $\mathbf{B}$  is defined as the  $nr \times pq$ -matrix

$$\mathbf{C} = \mathbf{A} \otimes \mathbf{B} = \begin{pmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \cdots & a_{1p}\mathbf{B} \\ \vdots & \vdots & & \vdots \\ a_{n1}\mathbf{B} & a_{n2}\mathbf{B} & \cdots & a_{np}\mathbf{B} \end{pmatrix}.$$

**Theorem A.4 Rules for the Kronecker Product**

Let  $A$ ,  $B$ ,  $C$ , and  $D$  be matrices of appropriate order and let  $k$  be a scalar. We then have the following rules:

1.  $k(A \otimes B) = (kA) \otimes B = A \otimes (kB)$
2.  $A \otimes (B \otimes C) = (A \otimes B) \otimes C$
3.  $A \otimes (B + C) = (A \otimes B) + (A \otimes C)$
4.  $(A \otimes B)' = A' \otimes B'$
5.  $(AB) \otimes (CD) = (A \otimes C)(B \otimes D)$

**Definition A.11 Orthogonal Matrices**

A square matrix  $A$  is called orthogonal if  $AA' = A'A = I$ .

**Theorem A.5 Properties of Orthogonal Matrices**

Let  $A$  be orthogonal. We then have:

1. Row and column vectors span an orthonormal basis, i.e., the vectors have (Euclidean) length 1 and are pairwise orthogonal.
2. If  $A$  and  $B$  are orthogonal, then  $AB$  is orthogonal.

**Definition A.12 Idempotent Matrices**

A square matrix  $A$  is called idempotent if  $AA = A^2 = A$ .

The  $n \times n$ -matrix

$$C := I_n - \frac{1}{n}\mathbf{1}\mathbf{1}',$$

a special idempotent matrix, is often used in statistics. The following holds:

1. Multiplication of  $C$  with an  $n \times 1$ -vector  $a$  gives

$$Ca = \begin{pmatrix} a_1 - \bar{a} \\ \vdots \\ a_n - \bar{a} \end{pmatrix},$$

i.e., one obtains the vector centered about the mean  $\bar{a}$  of its elements.

2. Multiplication of  $C$  with an  $n \times p$ -matrix  $A$  gives

$$CA = \begin{pmatrix} a_{11} - \bar{a}_1 & \cdots & a_{1p} - \bar{a}_p \\ \vdots & & \vdots \\ a_{n1} - \bar{a}_1 & \cdots & a_{np} - \bar{a}_p \end{pmatrix},$$

where  $\bar{a}_1, \dots, \bar{a}_p$  are the means of the columns of  $A$ .

3.  $C\mathbf{1} = \mathbf{0}$ .
4.  $\mathbf{1}'C = \mathbf{0}'$ .
5.  $\mathbf{1}\mathbf{1}'C = C\mathbf{1}\mathbf{1}' = \mathbf{0}$ .
6.  $\sum_{i=1}^n (x_i - \bar{x})^2 = \mathbf{x}'C\mathbf{x}$  where  $\mathbf{x} = (x_1, \dots, x_n)'$ .

### Theorem A.6 Properties of Idempotent Matrices

If  $A$  and  $B$  are idempotent, then:

1.  $AB = BA$ , and  $AB$  is also idempotent.
2.  $I - A$  is idempotent.
3.  $A(I - A) = (I - A)A = 0$ .

## A.2 Rank of a Matrix

### Definition A.13 Row Rank, Column Rank

Let  $A$  be of order  $n \times p$ . The column rank  $\text{rk}_c(A)$  of  $A$  is defined as the maximum number of linearly independent columns of  $A$ . Correspondingly, the row rank  $\text{rk}_r(A)$  is the maximum number of linearly independent rows of  $A$ .

### Theorem A.7

The column rank equals the row rank, i.e.,

$$\text{rk}_c(A) = \text{rk}_r(A).$$

### Definition A.14 Rank of a Matrix

The rank of an  $n \times p$ -matrix is defined as

$$\text{rk}(A) := \text{rk}_c(A) = \text{rk}_r(A) \leq \min\{n, p\}.$$

We denote  $\text{rk}(A)$  also by  $r(A)$ .

If  $\text{rk}(A) = \min\{n, p\}$ , then  $A$  has full rank and is called regular.

### Theorem A.8 Properties of Ranks

Let  $A$ ,  $B$ , and  $C$  be matrices of appropriate order. It follows:

1.  $\text{rk}(A) = \text{rk}(-A)$
2.  $\text{rk}(A') = \text{rk}(A)$
3.  $\text{rk}(A + B) \leq \text{rk}(A) + \text{rk}(B)$
4.  $\text{rk}(AB) \leq \min\{\text{rk}(A), \text{rk}(B)\}$
5.  $\text{rk}(I_n) = n$

### Definition A.15 Row Space, Column Space

The row space  $R(A)$  of an  $n \times p$ -matrix is the subspace of  $\mathbb{R}^n$  spanned by the rows of  $A$ , i.e.,

$$R(A) := \{x \in \mathbb{R}^n : x = Ay \text{ for some } y \in \mathbb{R}^p\}.$$

The column space is defined correspondingly.

### Definition A.16 Null Space

The null space  $N(A)$  of an  $n \times p$ -matrix  $A$  is defined as the set

$$N(A) := \{x \in \mathbb{R}^p : Ax = 0\}.$$

### Theorem A.9 Properties of the Null Space

Let  $A$  be an  $n \times p$ -matrix. We then have:

1. The null space is a subspace of  $\mathbb{R}^p$ .
2.  $\text{rk}(A) + \dim(N(A)) = p$ .
3.  $N(A'A) = N(A)$ .

### Definition A.17 Inverse of a Matrix

Let  $A$  be a square matrix. A matrix  $A^{-1}$  is called the inverse of  $A$  if

$$AA^{-1} = A^{-1}A = I.$$

### Theorem A.10 Existence and Uniqueness of the Inverse

The inverse  $A^{-1}$  of a square  $n \times n$ -matrix  $A$  exists if and only if  $\text{rk}(A) = n$ , i.e., if  $A$  is regular. The inverse is unique, and  $A$  is called invertible, regular, or nonsingular. If no inverse of  $A$  exists, it is called singular.

### Theorem A.11 Rules for Inverses

Let  $A$ ,  $B$ , and  $C$  be invertible matrices of the same order, and let  $k \neq 0$  be a scalar. The following rules then hold:

1.  $(A^{-1})^{-1} = A$
2.  $(kA)^{-1} = k^{-1}A^{-1} = \frac{1}{k}A^{-1}$
3.  $(A')^{-1} = (A^{-1})'$
4.  $(AB)^{-1} = B^{-1}A^{-1}$
5.  $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$
6. If  $A$  is symmetric,  $A^{-1}$  is also symmetric.
7. The inverse of a diagonal matrix  $A = \text{diag}(a_1, \dots, a_n)$  is

$$A^{-1} = \text{diag}(a_1^{-1}, \dots, a_n^{-1}).$$

8. If  $A$  is orthogonal, then  $A^{-1} = A'$ .

### A.3 Block Matrices and the Matrix Inversion Lemma

#### Definition A.18 Block Matrix, Block-Diagonal Matrix

A matrix  $A$  is called a block (or partitioned) matrix if it is partitioned into

$$A = \begin{bmatrix} A_1 & A_{12} & \dots & A_{1p} \\ A_{21} & A_2 & \dots & A_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ A_{p1} & \dots & \dots & A_p \end{bmatrix},$$

where  $A_1, \dots, A_p, A_{12}, \dots, A_{p1}, \dots$  are (non-overlapping) matrix sub-blocks of suitable dimensions. A special case, appearing several times in this book, is block-diagonal matrices

$$A = \begin{bmatrix} A_1 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & A_2 & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \dots & A_p \end{bmatrix},$$

where all off-diagonal matrices are  $\mathbf{0}$ . We often denote such block-diagonal matrices by

$$A = \text{blockdiag}(A_1, A_2, \dots, A_p).$$

#### Theorem A.12 Sums and Products of Block Matrices

Rules for sums and products of matrices can be generalized to block matrices. For example,

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} + \begin{bmatrix} E & F \\ G & H \end{bmatrix} = \begin{bmatrix} A + E & B + F \\ C + G & D + H \end{bmatrix}$$

and

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \cdot \begin{bmatrix} E & F \\ G & H \end{bmatrix} = \begin{bmatrix} AE + BG & AF + BH \\ CE + DG & CF + DH \end{bmatrix},$$

assuming that all dimensions of submatrices fit together appropriately.

#### Theorem A.13 Inversion of Block-Diagonal Matrices

Matrices can also be inverted blockwise. If all sub-blocks  $A_1, A_2, \dots, A_p$  in a block-diagonal matrix  $A = \text{blockdiag}(A_1, A_2, \dots, A_p)$  are quadratic and invertible, then



$$A^{-1} = \begin{bmatrix} A_1^{-1} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & A_2^{-1} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \dots & A_p^{-1} \end{bmatrix} = \text{blockdiag}(A_1^{-1}, A_2^{-1}, \dots, A_p^{-1}).$$

### Theorem A.14 Matrix Inversion Lemma

In general, blockwise inversion can be based on the following analytic inversion formula:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} & -A^{-1}B(D - CA^{-1}B)^{-1} \\ -(D - CA^{-1}B)^{-1}CA^{-1} & (D - CA^{-1}B)^{-1} \end{bmatrix},$$

where  $A$  and  $D$  are square matrices and  $A$  and  $(D - CA^{-1}B)^{-1}$  are nonsingular. Exchanging the roles of  $A$ ,  $B$  and  $C$ ,  $D$ , an analogous formula is

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} (A - BD^{-1}C)^{-1} & -(A - BD^{-1}C)^{-1}BD^{-1} \\ -D^{-1}C(A - BD^{-1}C)^{-1} & D^{-1} + D^{-1}C(A - BD^{-1}C)^{-1}BD^{-1} \end{bmatrix}.$$

Equating the left upper blocks gives

$$(A - BD^{-1}C)^{-1} = A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1}$$

which is the *matrix inversion lemma*. It appears in several variants in the literature. One of them is the expression

$$(A + B)^{-1} = A^{-1} - A^{-1}B(B + BA^{-1}B)^{-1}BA^{-1}$$

for the inverse of the sum of two matrices, where  $B$  is not necessarily nonsingular.

## A.4 Determinant and Trace of a Matrix

### Definition A.19 Determinant

The determinant of a square matrix  $A$  of order  $n \times n$  is defined as

$$|A| = \sum_{i=1}^n (-1)^{i+j} a_{ij} |A_{-ij}|,$$

where  $A_{-ij}$  is the  $((n-1) \times (n-1))$ -matrix obtained by deleting row  $i$  and column  $j$  from  $A$ . For scalar matrices  $A = (a_{11})$  we get  $|A| = a_{11}$ . For a  $2 \times 2$ -matrix,  $|A| = a_{11}a_{22} - a_{12}a_{21}$ .

### Theorem A.15 Determinant of the Transpose

Let  $A$  be a square matrix, then  $|A'| = |A|$ .

### Theorem A.16 Determinants of Some Matrices

Let  $A$  be a square matrix. It follows:

1. If a row (or column) of  $A$  consists of zeros only, then  $|A| = 0$ .
2. If  $A$  has two identical rows (columns), then  $|A| = 0$ .
3. Let  $A$  be a triangular matrix, i.e., all elements above or all elements below the main diagonal are zero. It follows that  $|A|$  is then the product of all diagonal elements.
4.  $|I| = 1$ .

### Theorem A.17 Properties of Determinants

1.  $|kA| = k^n |A|$
2.  $|A| \neq 0 \iff \text{rk}(A) = n$
3.  $|AB| = |A| \cdot |B|$
4.  $|A^{-1}| = \frac{1}{|A|}$
5. If  $A$  is orthogonal, then  $|A| = \pm 1$
6. Sylvester's theorem:  $|A + BC| = |A| |I + CA^{-1}B|$

### Definition A.20 Trace of a Matrix

Let  $A = (a_{ij})$  be an  $n \times n$ -matrix. The sum of the diagonal elements is then called trace of  $A$ , i.e.,

$$\text{tr}(A) = \sum_{i=1}^n a_{ii}.$$

### Theorem A.18 Properties of the Trace

The trace has the following properties:

1.  $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$ .
2.  $\text{tr}(A) = \text{tr}(A')$ .
3.  $\text{tr}(kA) = k \cdot \text{tr}(A)$ .
4.  $\text{tr}(AB) = \text{tr}(BA)$ . This remains valid if  $A$  is of order  $n \times p$  and  $B$  is of order  $p \times n$ .
5. Let  $x, y$  be vectors  $\in \mathbb{R}^n$ . It follows that  $\text{tr}(xy') = \text{tr}(yx') = \text{tr}(x'y) = x'y$ .

## A.5 Generalized Inverse

### Definition A.21 Generalized Inverse

Let  $A$  be an  $n \times p$ -matrix with  $n \leq p$ . The  $p \times n$ -matrix  $A^-$  is then called generalized inverse (or g-inverse) of  $A$  if

$$AA^-A = A.$$

### Theorem A.19 Existence of a Generalized Inverse

A generalized inverse always exists, but it is not unique in general.

### Theorem A.20 Properties of the Generalized Inverse

For any  $n \times p$ -matrix  $A$  and any  $p \times n$ -generalized inverse, we have

1.  $\text{rk}(A) = \text{rk}(AA^-) = \text{rk}(A^-A)$ .
2.  $\text{rk}(A) \leq \text{rk}(A^-)$ .
3. If  $A$  is regular, then  $A^- = A^{-1}$ , and  $A^{-1}$  is unique.
4.  $A^-A$  and  $AA^-$  are idempotent.

## A.6 Eigenvalues and Eigenvectors

### Definition A.22 Eigenvalue and Eigenvector

Let  $A$  be an  $n \times n$ -square matrix. The (possibly complex) number  $\lambda \in \mathbb{C}$  is called an eigenvalue of  $A$ , if there exists a nonzero vector  $x \in \mathbb{C}^n$  (with possibly complex elements) such that

$$Ax = \lambda x$$

or equivalently

$$(A - \lambda I)x = 0.$$

The vector  $x$  is then called a (right) eigenvector for the eigenvalues  $\lambda$ .

### Definition A.23 Characteristic Polynomials

Let  $A$  be an  $n \times n$ -square matrix; then

$$q(\lambda) := |A - \lambda I|$$

is called the characteristic polynomial of  $A$ .

Definition A.19 of the determinant of a matrix shows that  $q(\lambda)$  is indeed a polynomial of order  $n$ . Therefore,  $q(\lambda)$  has the form

$$q(\lambda) = (-\lambda)^n + \alpha_{n-1}(-\lambda)^{n-1} + \cdots + \alpha_1(-\lambda) + \alpha_0, \quad (\text{A.1})$$

where the scalar coefficients are unspecified. The polynomial can always be written as

$$q(\lambda) = |\mathbf{A} - \lambda \mathbf{I}| = \prod_{i=1}^n (\lambda_i - \lambda), \quad (\text{A.2})$$

where  $\lambda_1, \dots, \lambda_n$  are the  $n$  roots of the characteristic equation  $q(\lambda) = |\mathbf{A} - \lambda \mathbf{I}| = 0$ . The fundamental theorem of algebra states that  $q(\lambda) = 0$  has exactly  $n$  roots, which are not necessarily different and may be complex numbers.

### Theorem A.21 Computation of Eigenvalues via the Characteristic Polynomial

The eigenvalues  $\lambda_1, \dots, \lambda_n$  of a square matrix  $\mathbf{A}$  are the roots of the characteristic equation

$$|\mathbf{A} - \lambda \mathbf{I}| = 0.$$

### Theorem A.22 Properties of the Eigenvalues

The eigenvalues of an  $n \times n$ -matrix have the following properties:

1.  $|\mathbf{A}| = \prod_{i=1}^n \lambda_i$ .
2.  $\text{tr}(\mathbf{A}) = \sum_{i=1}^n \lambda_i$ .
3.  $\mathbf{A}$  is regular if and only if all eigenvalues are nonzero.
4. The matrices  $\mathbf{A}$  and  $\mathbf{A}'$  have the same eigenvalues.
5. If  $\lambda$  is an eigenvalue of a regular matrix  $\mathbf{A}$ , then  $\frac{1}{\lambda}$  is an eigenvalue of  $\mathbf{A}^{-1}$ .
6. The eigenvalues of a diagonal matrix are the elements of the diagonal.
7. The eigenvalues of an orthogonal matrix  $\mathbf{A}$  are either 1 or  $-1$ .
8. The eigenvalues of an idempotent matrix  $\mathbf{A}$  are either 1 or 0.

### Definition A.24 Eigenspace

Let  $\lambda$  be an eigenvalue of a square matrix  $\mathbf{A}$ . The set

$$\mathbf{A}_\lambda := \{\mathbf{x} \in \mathbb{C}^n | \mathbf{x} \text{ eigenvector for } \lambda\} \cup \{0\}$$

is called eigenspace for  $\lambda$ .

### Definition A.25 Similar Matrices

The matrices  $\mathbf{A}$  and  $\mathbf{B}$  are called similar (denoted as  $\mathbf{A} \sim \mathbf{B}$ ), if there exists a regular matrix  $\mathbf{C}$  such that  $\mathbf{B} = \mathbf{C} \mathbf{A} \mathbf{C}^{-1}$ .

### Theorem A.23 Eigenvalues of Similar Matrices

Let  $A$  and  $B$  be similar, then it follows:

1.  $A$  and  $B$  have the same characteristic polynomial and the same eigenvalues.
2. If  $x$  is an eigenvector of  $A$  for the eigenvalue  $\lambda$ , then  $Cx$  is an eigenvector of  $B = CAC^{-1}$ .

### Theorem A.24 Eigenvalues and Eigenvectors of Symmetric Matrices

Let  $A$  be a symmetric  $n \times n$ -matrix, then it follows:

1. All eigenvalues are real numbers.
2. Eigenvectors for different eigenvalues are pairwise orthogonal.

### Theorem A.25 Spectral Decomposition Theorem

Let  $A$  be a symmetric  $n \times n$ -matrix with rank  $\text{rk}(A) = r$ ; then  $A$  can be written as

$$A = P \text{diag}(\lambda_1, \dots, \lambda_r) P',$$

where  $\lambda_1, \dots, \lambda_r$  are the nonzero eigenvalues of  $A$  and  $P$  is an orthogonal  $n \times r$ -matrix whose columns are the corresponding orthonormal eigenvectors. Equivalently,

$$P'AP = \text{diag}(\lambda_1, \dots, \lambda_r).$$

### Theorem A.26 Spectral Decomposition of an Idempotent Matrix

Let the  $n \times n$ -matrix  $A$  be symmetric and idempotent with  $\text{rk}(A) = r$ ; then

$$P'AP = I_r,$$

and  $\text{rk}(A) = \text{tr}(A)$ .

## A.7 Quadratic Forms

### Definition A.26 Quadratic Form

Let  $A$  be a symmetric  $n \times n$ -matrix. A quadratic form in the vector  $x$  is a function of the form

$$Q(x) = x'Ax = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j = \sum_{i=1}^n a_{ii} x_i^2 + 2 \sum_{i=1}^n \sum_{j>i}^n a_{ij} x_i x_j.$$

### Definition A.27 Definite Matrices

The quadratic form  $x'Ax$  and the matrix  $A$  are called:

1. Positive definite (p.d.), if  $x'Ax > 0$  for all  $x \neq 0$ , notation:  $A > 0$
2. Positive semidefinite (p.s.d.), if  $x'Ax \geq 0$  and  $x'Ax = 0$  for at least one  $x \neq 0$

3. Nonnegative definite, if  $\mathbf{x}'\mathbf{A}\mathbf{x}$  and  $\mathbf{A}$  are either p.d. or p.s.d., notation:  $\mathbf{A} \geq 0$
4. Negative definite (n.d.), if  $-\mathbf{A}$  is positive definite
5. Negative semidefinite (n.s.d.), if  $-\mathbf{A}$  is p.s.d.
6. Indefinite in all other cases

### Theorem A.27 Criteria for Definite Matrices

Let  $\mathbf{A}$  be a symmetric matrix with real eigenvalues  $\lambda_1, \dots, \lambda_n$ . It then follows that  $\mathbf{A}$  is:

1. Positive definite, if and only if  $\lambda_i > 0$  for  $i = 1, \dots, n$
2. Positive semidefinite, if and only if  $\lambda_i \geq 0$  for  $i = 1, \dots, n$  and  $\lambda_i = 0$  for at least one eigenvalue
3. Negative definite, if and only if  $\lambda_i < 0$  for  $i = 1, \dots, n$
4. Negative semidefinite, if and only if  $\lambda_i \leq 0$  for  $i = 1, \dots, n$  and at least one  $\lambda_i = 0$
5. Indefinite, if and only if  $\mathbf{A}$  has at least one positive and one negative eigenvalue

### Theorem A.28 Properties of Positive Definite Matrices

For any positive definite matrix  $\mathbf{A}$  the following properties hold:

1.  $\mathbf{A}$  is regular and, thus, invertible.
2.  $\mathbf{A}^{-1}$  is positive definite.
3. The diagonal elements  $a_{ii}$ ,  $i = 1, \dots, n$ , are positive, i.e.  $a_{ii} > 0$ .
4.  $\text{tr}(\mathbf{A}) > 0$ .
5. If  $\mathbf{B}$  is positive semidefinite, then  $\mathbf{A} + \mathbf{B}$  is positive definite.

### Theorem A.29

Let  $\mathbf{A}$  be an  $n \times n$ -matrix and  $\mathbf{Q}$  an  $n \times m$ -matrix. The following then holds:

1. If  $\mathbf{A}$  is nonnegative definite, then  $\mathbf{Q}'\mathbf{A}\mathbf{Q}$  is nonnegative definite.
2. If  $\mathbf{A}$  is positive definite and if  $\text{rk}(\mathbf{Q}) = m$ , then  $\mathbf{Q}'\mathbf{A}\mathbf{Q}$  is positive definite.

### Theorem A.30

Let  $\mathbf{B}$  be an  $n \times p$ -matrix. The matrix  $\mathbf{B}'\mathbf{B}$  is then symmetric and nonnegative definite. If  $\text{rk}(\mathbf{B}) = p$ , then  $\mathbf{B}'\mathbf{B}$  is positive definite.

### Theorem A.31 Eigenvalues of $\mathbf{B}'\mathbf{B}$ and $\mathbf{B}\mathbf{B}'$

Let  $\mathbf{B}$  be an  $n \times p$ -matrix with  $\text{rk}(\mathbf{B}) = r$ . It follows that:

1.  $\mathbf{B}\mathbf{B}'$  and  $\mathbf{B}'\mathbf{B}$  have the same  $r$  positive eigenvalues  $\lambda_j$ ,  $j = 1, \dots, r$ .
2. If  $\mathbf{v}$  is an eigenvector of  $\mathbf{B}'\mathbf{B}$  for the eigenvalue  $\lambda$ , then

$$\mathbf{u} := \frac{1}{\sqrt{\lambda}}\mathbf{B}\mathbf{v}$$

is an eigenvector of  $\mathbf{B}\mathbf{B}'$  for  $\lambda$ .

### Theorem A.32 Cholesky Decomposition

Any symmetric and positive definite  $n \times n$ -matrix  $A$  can be uniquely decomposed into

$$A = LL',$$

where  $L$  is a lower triangular matrix with positive diagonal elements.  $L$  is called Cholesky factor of  $A$ .

## A.8 Differentiation of Matrix Functions

### Definition A.28 Differentiation with Respect to a Vector

Let  $\mathbf{x} = (x_1, \dots, x_n)'$  be a  $n \times 1$ -vector and  $f(\mathbf{x})$  a real function differentiable with respect to the elements  $x_i$  of  $\mathbf{x}$ .

The  $n \times 1$ -vector

$$\frac{\partial f}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$$

is then called differential of  $f$  with respect to  $\mathbf{x}$ . We denote by

$$\frac{\partial f}{\partial \mathbf{x}'} = \left( \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)$$

the transpose of  $\frac{\partial f}{\partial \mathbf{x}}$ .

To give an example, suppose that

$$f(\mathbf{x}) = \mathbf{y}'\mathbf{x} = \sum_{i=1}^n y_i x_i,$$

where  $\mathbf{y} = (y_1, \dots, y_n)'$  is constant. Then we have

$$\frac{\partial f}{\partial \mathbf{x}} = \frac{\partial \mathbf{y}'\mathbf{x}}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial (\sum y_i x_i)}{\partial x_1} \\ \frac{\partial (\sum y_i x_i)}{\partial x_2} \\ \vdots \\ \frac{\partial (\sum y_i x_i)}{\partial x_n} \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \mathbf{y}.$$

### Definition A.29 Differentiation of a Vector Function with Respect to a Vector

Let  $\mathbf{x} = (x_1, \dots, x_n)'$  be a  $n \times 1$ -vector and  $\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_m(\mathbf{x}))'$  a  $m \times 1$  vector function differentiable with respect to the elements  $x_i$  of  $\mathbf{x}$ .

The  $n \times m$ -matrix

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \left( \frac{\partial f_j}{\partial x_i} \right) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_1} \\ \vdots & & \vdots \\ \frac{\partial f_1}{\partial x_n} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}$$

is then called differential of  $\mathbf{f}$  with respect to  $\mathbf{x}$ . We denote by

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}'} = \left( \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right)' = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}$$

the transpose of  $\frac{\partial \mathbf{f}}{\partial \mathbf{x}}$ .

### Theorem A.33 Differentiation Rules

Assume that  $A$  is a matrix and  $\mathbf{a}$ ,  $\mathbf{x}$ , and  $\mathbf{y}$  are vectors. Furthermore, we assume that the following expressions exist and all matrices and vectors are of appropriate order. The following rules then hold:

1.  $\frac{\partial \mathbf{y}'\mathbf{x}}{\partial \mathbf{x}} = \mathbf{y}$ .
2.  $\frac{\partial \mathbf{x}'A\mathbf{x}}{\partial \mathbf{x}} = (A + A')\mathbf{x}$ .
3. If  $A$  is symmetric, then

$$\frac{\partial \mathbf{x}'A\mathbf{x}}{\partial \mathbf{x}} = 2A\mathbf{x} = 2A'\mathbf{x}.$$

4.  $\frac{\partial A\mathbf{x}}{\partial \mathbf{x}} = A'$ .
5.  $\frac{\partial A\mathbf{x}}{\partial \mathbf{x}'} = A$ .

### Theorem A.34 Local Extremes

Let  $\mathbf{x} = (x_1, \dots, x_n)'$  be a  $(n \times 1)$ -vector and  $f(\mathbf{x})$  a real function differentiable with respect to the elements  $x_i$  of  $\mathbf{x}$ . Define the vector



$$\mathbf{s}(\mathbf{x}) = \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}}$$

of first derivatives and the matrix

$$\mathbf{H}(\mathbf{x}) = \frac{\partial \mathbf{s}(\mathbf{x})}{\partial \mathbf{x}'} = \begin{pmatrix} \frac{\partial s_1(\mathbf{x})}{\partial x_1} & \cdots & \frac{\partial s_1(\mathbf{x})}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial s_n(\mathbf{x})}{\partial x_1} & \cdots & \frac{\partial s_n(\mathbf{x})}{\partial x_n} \end{pmatrix} = \begin{pmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_n} \end{pmatrix}$$

of second derivatives.  $\mathbf{H}(\mathbf{x})$  is also called *Hessian matrix* or simply the *Hessian*.

A necessary condition for  $\mathbf{x} = \mathbf{x}_0$  being a local extreme of  $f$  is

$$\mathbf{s}(\mathbf{x}_0) = \mathbf{0}. \quad (\text{A.3})$$

If Eq. (A.3) is true, the following sufficient condition holds:

- If  $\mathbf{H}(\mathbf{x}_0)$  is positive definite  $\mathbf{x}_0$  is a local minimum.
- If  $\mathbf{H}(\mathbf{x}_0)$  is negative definite  $\mathbf{x}_0$  is a local maximum.